CHARMM-GUI
Effective Simulation Input Generator and More

CHARMM is a versatile program for atomic-level simulation of many-particle systems, particularly macromolecules of biological interest. - M. Karplus

Some lectures, job postings, and FAQ are now available. See update log for update history and giving for donation. Contact info is given below. Logout

Membrane Builder

User Profile

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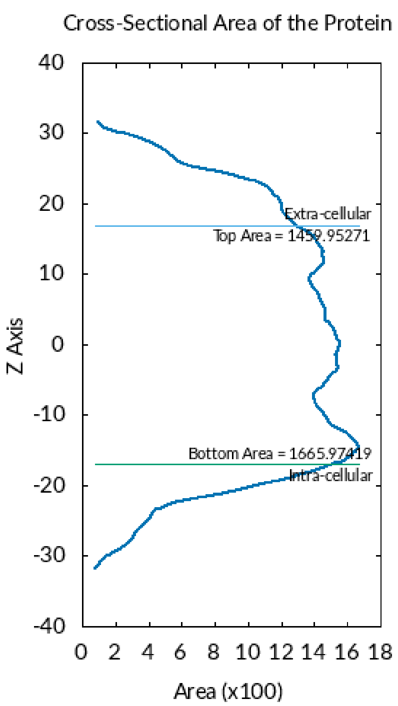
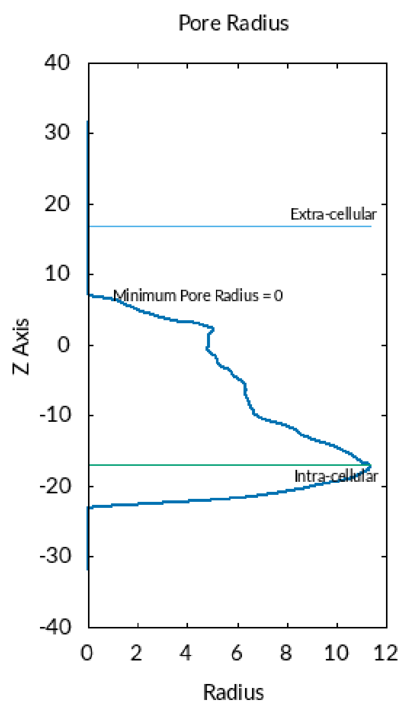
PDB Info | STEP 1 | STEP 2 | STEP 3 | STEP 4 | STEP 5 | STEP 6 JOB ID: 3750189754

download.tgz

CHARMM PDB:	step1_pdbreader.pdb (view structure)	
Orientation Input:	step2_orient.inp	
Orientation Output:	step2_orient.out	
Oriented PDB:	step2_orient.pdb (view structure) (please view this structure before you move to the next step)	
Area Calculation:	step2_area.str	Calculate cross sectional area of the protein
	step2_area.plo	Computed cross sectional area along Z axis
	step2_protein_area.str	Top/Bottom area of the protein
Pore Water:	step2.1_pore.inp	
	step2.1_pore.out	
	step2.1_pore.pdb (view structure)	
	step2.1_porewat.pdb (view structure)	
	step2.1_porewat.crd	

Calculated Cross Sectional Area:





Protein Top Area: 1459.95271
Protein Bot Area: 1665.97419
X Extent: -31 to 23
Y Extent: -24 to 26
Z Extent: -33 to 32
Pore Minimum Radius: 0

Protein Projection onto XY:

Projection of protein portions above and below the membrane surface onto the membrane surface can be performed only to exclude pseudo lipid spheres underneath the protein portion during the lipid sphere packing.

☐ Turn on protein projection onto XY of upper leaflet
☐ Turn on protein projection onto XY of lower leaflet
This option is not recommended to use unless it is absolutely necessary

System Size Determination Options:

○ **Homogeneous Lipid** **"Homogeneous Lipid" option is no longer supported. You can use "Heterogeneous Lipid" option even for homogeneous lipid bilayer building.**

Heterogeneous Lipid

1. Box Type: Rectangular (Currently, only CHARMM, NAMD, and GROMACS support the hexagonal box)

2. Length of Z based on:
- Water thickness 15 (Minimum water height on top and bottom of the system)
- Hydration number 50 (Number of water molecules per one lipid molecule)

3. Length of XY based on:
- Ratios of lipid components
- Numbers of lipid components

XY Dimension Ratio: 1
(The system size along the X and Y must be the same)

Show the system info click this once you fill the following table:

Lipid Type	Charge [e]	Tail Info. [sn1/sn2]	Images	# of Lipid on Upperleaflet	# of Lipid on Lowerleaflet	Surface Area
▸ **Sterols**						
▸ **PA (phosphatidic acid) Lipids**						
▾ **PC (phosphatidylcholine) Lipids**						
DDPC	0	10:0 / 10:0	[Image]	0	0	64.1
DCPC	0	11:0 / 11:0	[Image]	0	0	64.1
DLPC	0	12:0 / 12:0	[Image]	0	0	64.1
DMPC	0	14:0 / 14:0	[Image]	0	0	64.1
DPPC	0	16:0 / 16:0	[Image]	0	0	63.0
DSPC	0	18:0 / 18:0	[Image]	0	0	65.6
PSPC	0	16:0 / 18:0	[Image]	0	0	68.3
PYPC	0	16:0 / 16:1	[Image]	0	0	64.1
POPC	0	16:0 / 18:1	[Image]	107	104	68.3
PLPC	0	16:0 / 18:2	[Image]	0	0	65.2
PiLPC	0	16:0 / 18:2	[Image]	0	0	60.0
PEPC	0	16:0 / 22:1	[Image]	0	0	66.5
SOPC	0	18:0 / 18:1	[Image]	0	0	66.2
SLPC	0	18:0 / 18:2	[Image]	0	0	65.2
DRPC	0	14:1 / 14:1	[Image]	0	0	67.0
DYPC	0	16:1 / 16:1	[Image]	0	0	67.0
YOPC	0	16:1 / 18:1	[Image]	0	0	67.0
DOPC	0	18:1 / 18:1	[Image]	0	0	69.7
OEPC	0	18:1 / 22:1	[Image]	0	0	66.5
DUPC	0	18:2 / 18:2	[Image]	0	0	70.0
DGPC	0	20:1 / 20:1	[Image]	0	0	66.7
DEPC	0	22:1 / 22:1	[Image]	0	0	66.5
DNPC	0	24:1 / 24:1	[Image]	0	0	62.3
DLiPC	0	18:2 / 18:2	[Image]	0	0	65.2

- ▸ **PE (phosphatidylethanolamine) Lipids**
- ▸ **PG (phosphatidylglycerol) Lipids**
- ▸ **PS (phosphatidylserine) Lipids**
- ▸ **PP (pyrophosphate) Lipids**
- ▸ **PI (phosphatidylinositol) Lipids**
- ▸ **CL (cardiolipin) Lipids**
- ▸ **DAG (diacylglycerol) Lipids** ☐ **Optimized charges for DAG**
- ▸ **TAG (triacylglycerol) Lipids** ☐ **Optimized charges for TAG**
- ▸ **TAP (trimethylammonium propane) & DAP/DMA (dimethylamino) Lipids**
- ▸ **PUFA (polyunsaturated fatty acid) Esterified Lipids**
- ▸ **SM (sphingo) and Ceramide Lipids**
- ▸ **Ether/Plasmalogen Lipids**
- ▸ **Bacterial Lipids**
- ▸ **Endosomal Lipids**
- ▸ **Ubiquinone/Ubiquinol & Archaeal Lipids**
- ▸ **Oxidized Lipids** ?

Calculated XY System Size:

	Upperleaflet	Lowerleaflet
Protein Area	1459.95271	1665.97419
Lipid Area	7548.1	7343.2
# of Lipids	115	112
Total Area	9008.05271	9009.17419
Protein X Extent	30.20	
Protein Y Extent	25.70	
Average Area	9008.61	
A	94.91	
B	94.91	

▼ **Fatty Acids**

LAU	-1	12:0	[Image]	0	0	30.0
MYR	-1	14:0	[Image]	0	0	30.0
PAL	-1	16:0	[Image]	0	0	30.0
STE	-1	18:0	[Image]	0	0	30.0
ARA	-1	20:0	[Image]	0	0	30.0
BEH	-1	22:0	[Image]	0	0	30.0
TRI	-1	23:0	[Image]	0	0	30.0
LIGN	-1	24:0	[Image]	0	0	30.0
MYRO	-1	14:1 (Δ^{9})	[Image]	0	0	30.0
PALO	-1	16:1 (Δ^{9})	[Image]	0	0	30.0
HTA	-1	16:3 ($\Delta^{7,10,13}$)	[Image]	0	0	30.0
OLE	-1	18:1 (Δ^{9})	[Image]	0	0	30.0
LIN	-1	18:2 ($\Delta^{9,12}$)	[Image]	0	0	30.0
ALIN	-1	18:3 ($\Delta^{9,12,15}$)	[Image]	0	0	30.0
SDA	-1	18:4 ($\Delta^{6,9,12,15}$)	[Image]	0	0	30.0
GLA	-1	18:3 ($\Delta^{6,9,12}$)	[Image]	0	0	30.0
EICO	-1	20:1 (Δ^{11})	[Image]	0	0	30.0
EDA	-1	20:2 ($\Delta^{11,14}$)	[Image]	0	0	30.0
MEA	-1	20:3 ($\Delta^{5,8,11}$)	[Image]	0	0	30.0
DGLA	-1	20:3 ($\Delta^{8,11,14}$)	[Image]	0	0	30.0
ETE	-1	20:3 ($\Delta^{11,14,17}$)	[Image]	0	0	30.0
ETA	-1	20:4 ($\Delta^{8,11,14,17}$)	[Image]	0	0	30.0
EPA	-1	20:5 ($\Delta^{5,8,11,14,17}$)	[Image]	0	0	30.0
ARAN	-1	20:4 ($\Delta^{5,8,11,14}$)	[Image]	8	8	30.0
HPA	-1	21:5 ($\Delta^{6,9,12,15,18}$)	[Image]	0	0	30.0
ERU	-1	22:1 (Δ^{13})	[Image]	0	0	30.0
DDA	-1	22:2 ($\Delta^{13,16}$)	[Image]	0	0	30.0
ADR	-1	22:4 ($\Delta^{7,10,13,16}$)	[Image]	0	0	30.0
DPT	-1	22:5 ($\Delta^{4,7,10,13,16}$)	[Image]	0	0	30.0
DPA	-1	22:5 ($\Delta^{7,10,13,16,19}$)	[Image]	0	0	30.0
DHA	-1	22:6 ($\Delta^{4,7,10,13,16,19}$)	[Image]	0	0	30.0
NER	-1	24:1 (Δ^{15})	[Image]	0	0	30.0
TTA	-1	24:4 ($\Delta^{9,12,15,18}$)	[Image]	0	0	30.0
TPT	-1	24:5 ($\Delta^{6,9,12,15,18}$)	[Image]	0	0	30.0
TPA	-1	24:5 ($\Delta^{9,12,15,18,21}$)	[Image]	0	0	30.0
THA	-1	24:6 ($\Delta^{6,9,12,15,18,21}$)	[Image]	0	0	30.0
LAUP	0	12:0	[Image]	0	0	30.0
MYRP	0	14:0	[Image]	0	0	30.0
PALP	0	16:0	[Image]	0	0	30.0
STEP	0	18:0	[Image]	0	0	30.0
ARAP	0	20:0	[Image]	0	0	30.0
BEHP	0	22:0	[Image]	0	0	30.0
TRIP	0	23:0	[Image]	0	0	30.0
LIGNP	0	24:0	[Image]	0	0	30.0
MYROP	0	14:1 (Δ^{9})	[Image]	0	0	30.0
PALOP	0	16:1 (Δ^{9})	[Image]	0	0	30.0
HTAP	0	16:3 ($\Delta^{7,10,13}$)	[Image]	0	0	30.0
OLEP	0	18:1 (Δ^{9})	[Image]	0	0	30.0
LINP	0	18:2 ($\Delta^{9,12}$)	[Image]	0	0	30.0
ALINP	0	18:3 ($\Delta^{9,12,15}$)	[Image]	0	0	30.0
SDAP	0	18:4 ($\Delta^{6,9,12,15}$)	[Image]	0	0	30.0
GLAP	0	18:3 ($\Delta^{6,9,12}$)	[Image]	0	0	30.0
EICOP	0	20:1 (Δ^{11})	[Image]	0	0	30.0
EDAP	0	20:2 ($\Delta^{11,14}$)	[Image]	0	0	30.0
MEAP	0	20:3 ($\Delta^{5,8,11}$)	[Image]	0	0	30.0
DGLAP	0	20:3 ($\Delta^{8,11,14}$)	[Image]	0	0	30.0
ETEP	0	20:3 ($\Delta^{11,14,17}$)	[Image]	0	0	30.0
ETAP	0	20:4 ($\Delta^{8,11,14,17}$)	[Image]	0	0	30.0
EPAP	0	20:5 ($\Delta^{5,8,11,14,17}$)	[Image]	0	0	30.0
ARANP	0	20:4 ($\Delta^{5,8,11,14}$)	[Image]	0	0	30.0
HPAP	0	21:5 ($\Delta^{6,9,12,15,18}$)	[Image]	0	0	30.0
ERUP	0	22:1 (Δ^{13})	[Image]	0	0	30.0
DDAP	0	22:2 ($\Delta^{13,16}$)	[Image]	0	0	30.0
ADRP	0	22:4 ($\Delta^{7,10,13,16}$)	[Image]	0	0	30.0
DPTP	0	22:5 ($\Delta^{4,7,10,13,16}$)	[Image]	0	0	30.0
DPAP	0	22:5 ($\Delta^{7,10,13,16,19}$)	[Image]	0	0	30.0

DHAP	0	22:6 ($\Delta^{4,7,10,13,16,19}$)	[Image]	0	0	30.0
NERP	0	24:1 (Δ^{15})	[Image]	0	0	30.0
TTAP	0	24:4 ($\Delta^{9,12,15,18}$)	[Image]	0	0	30.0
TPTP	0	24:5 ($\Delta^{6,9,12,15,18}$)	[Image]	0	0	30.0
TPAP	0	24:5 ($\Delta^{9,12,15,18,21}$)	[Image]	0	0	30.0
THAP	0	24:6 ($\Delta^{6,9,12,15,18,21}$)	[Image]	0	0	30.0

▶ **Detergents**

▶ **N-acylated Amino Acids**

▶ **PEG (polyethylene glycol) Lipids**

▶ **Glycolipids**

▶ **LPS (lipopolysaccharides)**

Note: The surface areas listed above are only suggested values. In the case of high cholesterol concentration, 10~20% smaller surface areas will be adopted as default values, which might be more reasonable and close to equilibrium lipid areas. Please refer the manuscript or the following link to list of default surface area we are using. ?

Next Step:
Determine the System Size